ETF Target Weight Optimization

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Maximizing Sharpe Ratio with Analyst-Supplied Expected Returns

1. Overview

This analysis optimizes allocation weights across seven ETFs by maximizing the Sharpe ratio. Rather than relying on noisy historical mean returns, the model uses analyst-supplied forward-looking return assumptions combined with a historical covariance matrix to determine target weights.

Design Choice	Detail
Expected Returns	Analyst-supplied forward-looking views — not historical means
Covariance Matrix	Historical daily log returns × 252 (2021–present)
Risk-Free Rate	5.00%
Objective	Maximize Sharpe Ratio via SLSQP solver
Position Bounds	Minimum 4% | Maximum 24% per ETF
Benchmark	Equal-weight 1/N portfolio using the same analyst μ

2. Asset Universe & Return Assumptions

Seven ETFs spanning equities, fixed income, and inflation protection were selected. All expected returns are analyst-supplied forward-looking views.

Ticker	Asset Class	Analyst μ	Hist. Vol	Implied Sharpe
IWV	US Equity (Russell 3000)	**8.00%**	~14%	**0.214**
IEMG	Emerging Market Equity	**8.00%**	~17%	**0.176**
VEA	Developed Non-US Equity	**7.90%**	~14%	**0.207**
JNK	High-Yield Bond	**6.60%**	~9%	**0.178**
SPTL	Long-Term Government Bond	**5.10%**	~15%	**0.007**
VCIT	Investment-Grade Corp Bond	**4.90%**	~7%	**-0.014**
TIP	Inflation-Protected Bond (TIPS)	**4.70%**	~8%	**-0.038**

Implied Sharpe = (μ − Rf) / σ | Rf = 5.00% | Ignore cross-asset correlations.

3. Portfolio Performance Results

The Max-Sharpe optimizer delivered a 33.7% improvement in Sharpe Ratio over the equal-weight baseline, with higher expected return and lower volatility.

Metric	Equal-Weight	Max-Sharpe Opt.	Δ Change
Expected Return	6.64%	**7.11%**	**+0.47%**
Portfolio Volatility	9.32%	**8.82%**	**−0.50%**
Sharpe Ratio	0.178	**0.238**	**+0.060**

Key insight: the optimizer simultaneously raised return (+47bp) and reduced risk (−50bp), delivering a superior risk-adjusted outcome. The Sharpe ratio improvement (+34%) is attributable entirely to weight allocation, not different data inputs.

4. Optimal Portfolio Weights

The SLSQP optimizer converged in under 100 iterations. All three equity ETFs hit the maximum bound (24%); SPTL and TIP hit the minimum bound (4%).

Ticker	Asset Class	Equal Wt.	Opt. Weight	Δ vs Equal	Flag
IWV	US Equity (Russell 3000)	14.3%	**24.0%**	**+9.7%**	**MAX ▲**
IEMG	Emerging Market Equity	14.3%	**24.0%**	**+9.7%**	**MAX ▲**
VEA	Developed Non-US Equity	14.3%	**24.0%**	**+9.7%**	**MAX ▲**
JNK	High-Yield Bond	14.3%	**10.0%**	**−4.3%**	**FREE**
SPTL	Long-Term Government Bond	14.3%	**4.0%**	**−10.3%**	**MIN ▼**
VCIT	Investment-Grade Corp Bond	14.3%	**10.0%**	**−4.3%**	**FREE**
TIP	Inflation-Protected Bond (TIPS)	14.3%	**4.0%**	**−10.3%**	**MIN ▼**

MAX ▲ = optimizer would allocate more if unconstrained (primary Sharpe driver).

MIN ▼ = optimizer would exclude if unconstrained (poor marginal risk/return contribution).

FREE = internal solution.

5. Efficient Frontier

The full efficient frontier was traced by solving the minimum-variance problem across 100 target return levels using the same analyst μ and historical Σ.

Portfolio	Return	Volatility	Sharpe
Global Minimum Variance	5.18%	7.21%	**0.025**
Tangency (Max Sharpe)	7.11%	8.82%	**0.238**
Equal-Weight (1/N)	6.64%	9.32%	**0.178**

The tangency portfolio (Max-Sharpe) lies on the Capital Market Line — the single point on the efficient frontier that maximizes risk-adjusted return given the 5% risk-free rate.